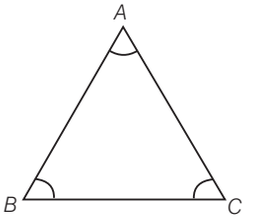
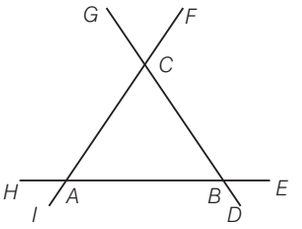
12.3 Triangles

Triangle : A three sided closed plane figure, which is formed by joining the three non-collinear points, is called as a triangle. It is denoted by the symbol Δ.



In the above Δ (triangle) ABC, A, B and C are three **vertices**, line segments AB, BC and AC are the three **sides** of the triangle. $\angle A, \angle B$ and $\angle C$ are the three **interior angles** of a triangle ABC.

In the adjoining figure $\angle FCB$, $\angle CBE$, $\angle ABD$, $\angle IAB$, $\angle HAC$, $\angle GCA$ are the **exterior angles** of the ΔABC.



Sum of the three interior angles of a triangle is always 180°.

Exterior angle = Sum of two interior opposite angles

e.g., $\angle CBE = \angle CAB + \angle BCA$

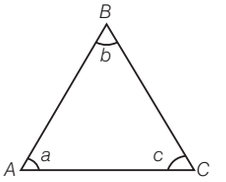
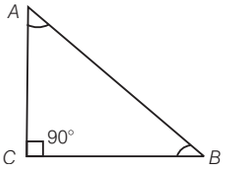
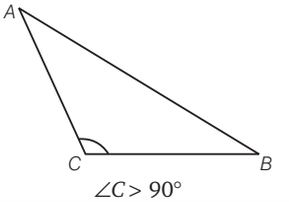
Perimeter of triangle is equal to sum of all the three sides *i.e.,* $a + b + c$

Semiperimeter of a triangle is half of the perimeter

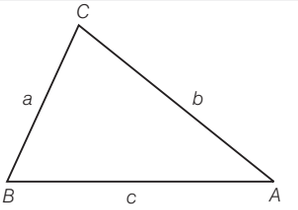
i.e., $s = \frac{a+b+c}{2}$, a, b, c are the length of three sides of a triangle.

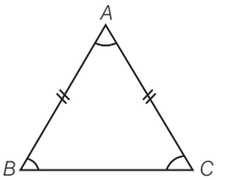
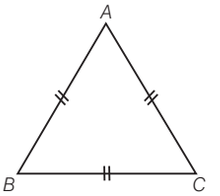
Types of Triangles

(A) According to interior angles

Types of Triangles	Property/Definition	Diagram
Acute angle triangle	Each of the angle of a triangle is less than 90° *i.e.,* $a < 90°, b < 90°, c < 90°$	 $\{\angle a, \angle b, \angle c\} < 90°$
Right angled triangle	One of the angle is equal to 90°, then it is called as right angled triangle. Rest two angles are complementary to each other.	 $\angle C = 90°$
Obtuse angle triangle	One of the angle is obtuse (*i.e.,* greater than 90°), then it is called as obtuse angle triangle.	 $\angle C > 90°$

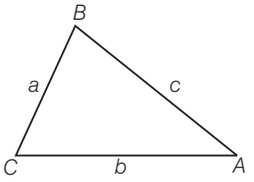
(B) According to the length of sides.

Types of Triangles	Property/Definition	Diagram
Scalene triangle	A triangle in which none of the three sides is equal is called a scalene triangle (all the three angles are also different).	 $a \neq b \neq c$

Isosceles triangle	A triangles in which at least two sides are equal is called an isosceles triangle. In this triangle, the angles opposite to the congruent sides are also equal.	 $AB = AC,\ \angle B = \angle C$
Equilateral triangle	A triangle in which all the three sides are equal called an equilateral triangle. In this triangle each angle is congruent and equal to 60°.	 $AB = BC = AC$ $\angle A = \angle B = \angle C = 60°$

Fundamental Properties of Triangles

1. Sum of any two sides is always greater than the third side.
2. The difference of any two sides is always less than the third side.
3. Greater angle has a greater side opposite to it and smaller angle has a smaller side opposite to it *i.e.,* if two sides of a triangle are not congruent then the angle opposite to the greater side is greater.
4. Let a, b and c be the three sides of a ΔABC and c is the largest side. Then



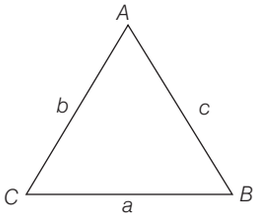
   (i) if $c^2 < a^2 + b^2$, the triangle is **acute angle triangle**

   (ii) if $c^2 = a^2 + b^2$, the triangle is **right angled triangle**

   (iii) if $c^2 > a^2 + b^2$, the triangle is **obtuse angle triangle**

5. **Sine rule :** In a ΔABC, if a, b, c be the three sides opposite to the angles A, B, C respectively, then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



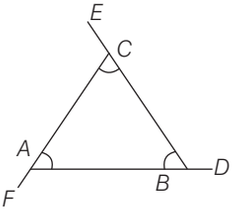
6. **Cosine rule :** In a ΔABC, if a, b, c be the sides opposite to angle A, B and C respectively, then

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \cos B = \frac{c^2 + a^2 - b^2}{2ca},$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

   (These rules have been discussed already in trigonometry.)

7. The sum of all the three interior angles is always 180°



   i.e., $\angle CAB + \angle ABC + \angle BCA = 180°$

8. The sum of three (ordered) exterior angles of a triangle is 360°

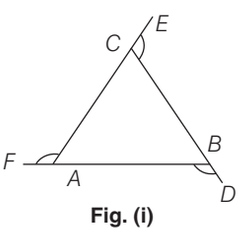


Fig. (i)

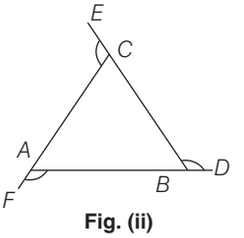


Fig. (ii)

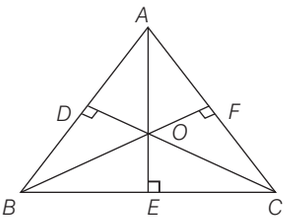
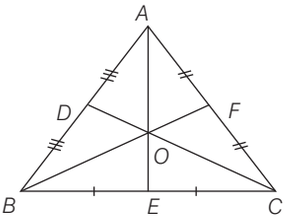
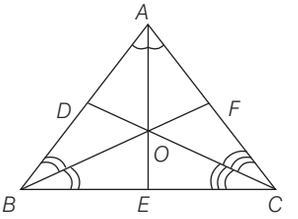
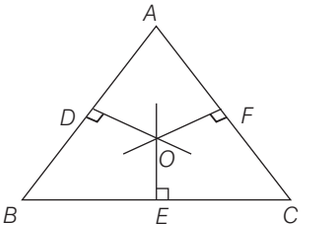
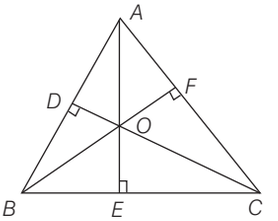
   In fig. (i) : $(\angle FAC + \angle ECB + \angle DBA) = 360°$

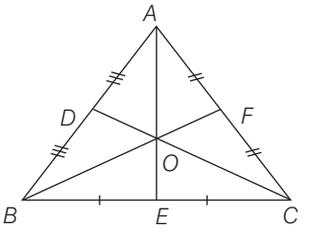
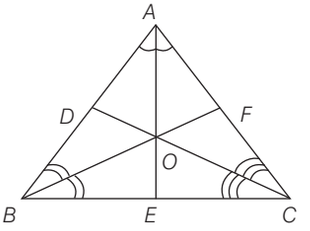
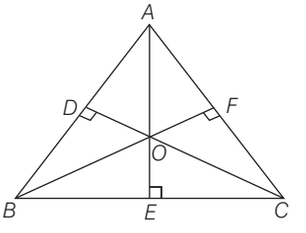
   In fig. (ii) : $(\angle FAB + \angle DBC + \angle ECA) = 360°$

9. The sum of an interior angle and its adjacent exterior angle is 180°.

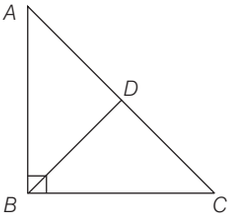
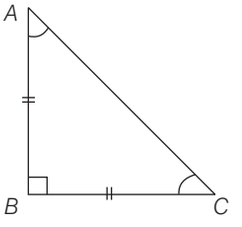
10. A triangle must has at least two acute angles.
11. In a triangle, the measure of an exterior angle equals the sum of the measures of the interior opposite angles.
12. The measure of an exterior angle of a triangle is greater than the measure of each of the opposite interior angles.

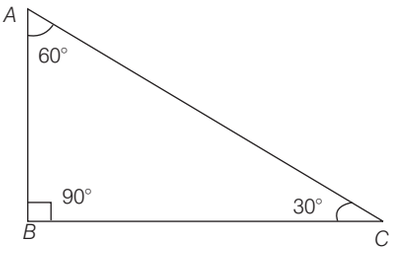
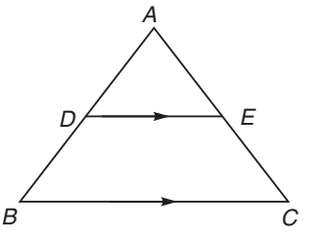
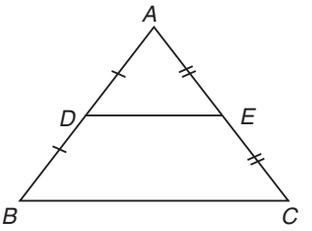
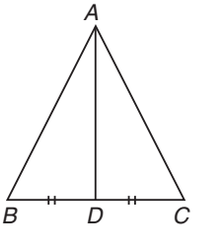
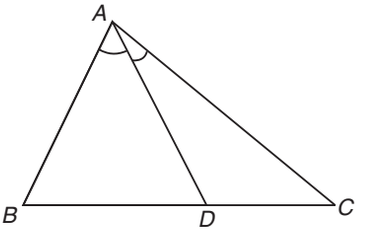
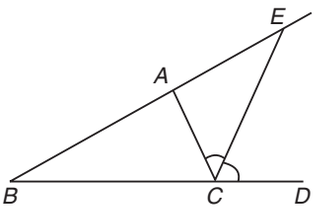
Important Definitions

Nomenclature	Property/Definition	Diagram
Altitude (or height)	The perpendicular drawn from the opposite vertex of a side in a triangle is called an altitude of the triangle. • There are three altitudes in a triangle.	 *AE*, *CD* and *BF* are the altitudes
Median	The line segment joining the mid-point of a side to the vertex opposite to the side is called a median. • There are three medians in a triangle. • A median bisects the area of the triangle *i.e.,* $A(\Delta ABE) = A(\Delta AEC) = \frac{1}{2} A(\Delta ABC)$ etc.	 *AE*, *CD* and *BF* are the medians ($BE = CE$, $AD = BD$, $AF = CF$)
Angle bisector	A line segment which originates from a vertex and bisects the same angle is called an angle bisector. $\left(\angle BAE = \angle CAE = \frac{1}{2} \angle BAC\right)$ etc.	 *AE*, *CD* and *BF* are the angle bisectors.
Perpendicular bisector	A line segment which bisects a side perpendicularly (i.e, at right angle) is called a perpendicular bisector of a side of triangle. • All points on the perpendicular bisector of a line are equidistant from the ends of the line.	 *DO*, *EO* and *FO* are the perpendicular bisectors.
Orthocentre	The point of intersection of the three altitudes of the triangle is called the orthocentre. $\angle BOC = 180 - \angle A$ $\angle COA = 180 - \angle B$ $\angle AOB = 180 - \angle C$	 '*O*' is the orthocentre

Types of Triangles	Property/Definition	Diagram
Centroid	The point of intersection of the three medians of a triangle is called the centroid. A centroid divides each median in the ratio 2 : 1 (vertex : base) $\frac{AO}{OE}=\frac{CO}{OD}=\frac{BO}{OF}=\frac{2}{1}$	 '*O*' is the centroid.
Incentre	The point of intersection of the angle bisectors of a triangle is called the incentre. Incentre *O* is the always equidistant from all three sides *i.e.,* the perpendicular distance between the sides and incentre is always same for all the three sides.	 '*O*' is the incentre.
Circumcentre	The point of intersection of the perpendicular bisectors of the sides of a triangle is called the circumcentre. $OA = OB = OC$ = (circum radius) Circumcentre *O* is always equidistant from all the three vertices *A, B* and *C*.	 '*O*' is the circumcentre.

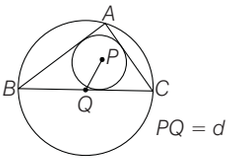
Important Theorems on Triangles

Theorem	Statement/Explanation	Diagram
Pythagoras theorem	The square of the hypotenuse of a right angled triangle is equal to the sum of the squares of the other two sides. *i.e.,* $(AC)^2 = (AB)^2 + (BC)^2$ • The converse of this theorem is also true. • The numbers which satisfy this relation, are called Pythagorean triplets. *e.g.,* (3, 4, 5), (5, 12, 13), (7, 24, 25), (8, 15, 17), (9, 40, 41), (11, 60, 61), (12, 35, 37), (16, 63, 65), (20, 21, 29), (28, 45, 53), (33, 56, 65) **Note :** All the multiples (or submultiples) of Pythagorean triplets also satisfy the relation. *e.g.,* (6, 8,10), (15,36, 39), (1.5, 2, 2.5) etc	 $\angle B = 90°$ $\overline{AC} \rightarrow$ Hypotenuse $AD = CD = BD$ (*D* is the mid-point of *AC*)
45° – 45° – 90° triangle theorem	If the angles of a triangle are 45°, 45° and 90°, then the hypotenuse (*i.e.,* longest side) is $\sqrt{2}$ times of any smaller side. Excluding hypotenuse rest two sides are equal. *i.e., AB = BC* and $AC = \sqrt{2}AB = \sqrt{2}BC$	 $\angle A = 45°, \angle B = 90°, \angle C = 45°$

Theorem	Statement/Explanation	Diagram
30° – 60° – 90° triangle theorem	If the angles of a triangle are 30°, 60° and 90°,then the sides opposite to 30° angle is half of the hypotenuse and the side opposite to 60° is $\frac{\sqrt{3}}{2}$ times the hypotenuse . e.g., $AB = \frac{AC}{2}$ and $BC = \frac{\sqrt{3}}{2}AC$ $\therefore \quad AB : BC : AC = 1 : \sqrt{3} : 2$	
Basic proportionality theorem (BPT) **or Thales theorem**	Any line parallel to one side of a triangle divides the other two sides proportionally. So if DE is drawn parallel to BC, it would divide sides AB and AC proportionally *i.e.,* $\frac{AD}{DB} = \frac{AE}{EC}$ or $\frac{AD}{AB} = \frac{AE}{AC}$ $\frac{AD}{DE} = \frac{AB}{BC} = \frac{AE}{DE} = \frac{AC}{BC}$	
Mid-point theorem	If the mid-points of two adjacent sides of a triangle are joined by a line segment, then this segment is parallel to the third side. *i.e.,* if $AD = BD$ and $AE = CE$, then $DE \parallel BC$	
Apollonius theorem	In a triangle, the sum of the squares of any two sides of a triangle is equal to twice the sum of the square of the median to the third side and square of half the third side. *i.e.,* $AB^2 + AC^2 = 2(AD^2 + BD^2)$	 $BD = CD$ AD is the median
Interior angle bisector theorem	In a triangle the angle bisector of an angle divides the opposite side to the angle in the ratio of the remaining two sides. *i.e.,* $\frac{BD}{CD} = \frac{AB}{AC}$ and $BD \times AC - CD \times AB = AD^2$	
Exterior angle bisector theorem	In a triangle the angle bisector of any exterior angle of a triangle divides the side opposite to the external angle in the ratio of the remaining two sides *i.e.,* $\frac{BE}{AE} = \frac{BC}{AC}$	

Euler's Theorem for a triangle

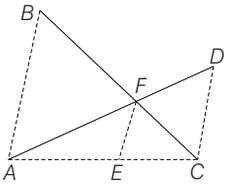
Let ΔABC have circumradius R and inradius r. Let d be the distance between the circumcentre and the incenter. Then we have $d^2 = R(R - 2r)$



Crossed Ladder Theorem

Let the two line segments BC and AD intersect at a point F, such that the point E lies on AC and $AB \parallel CD \parallel EF$. Then, we have

$$\frac{1}{AB} + \frac{1}{CD} = \frac{1}{EF}$$

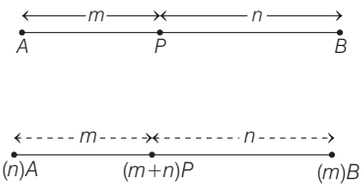


Mass Point Geometry

Let us consider a line AB which is intersected by a point P, such that $AP = m$ and $BP = n$, we have

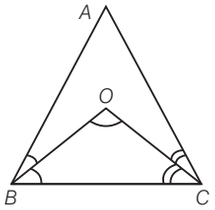
$$\frac{\text{wt}(A)}{\text{wt}(B)} = \frac{n}{m}$$

Therefore, the weight at A will be n, the weight at B will be m and the total weight at P will be $(m + n)$.

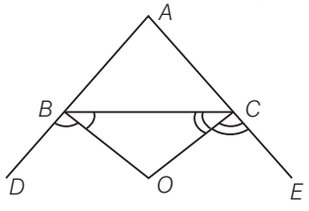


Useful Results

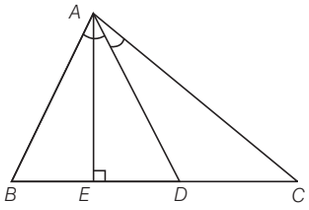
1. In a ΔABC, if the bisectors of $\angle B$ and $\angle C$ meet at O, then $\angle BOC = 90° + \frac{1}{2}\angle A$



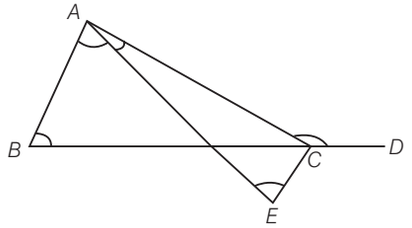
2. In a ΔABC, if sides AB and AC are produced to D and E respectively and the bisectors of $\angle DBC$ and $\angle ECB$ intersect at O, then $\angle BOC = 90° - \frac{1}{2}\angle A$



3. In a ΔABC, if AD is the angle bisector of $\angle BAC$ and $AE \perp BC$, then $\angle DAE = \frac{1}{2}(\angle ABC - \angle ACB)$

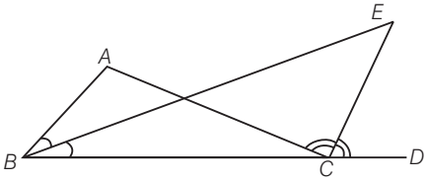


4. In a ΔABC, if BC is produced to D and AE is the angle bisector of $\angle A$, then



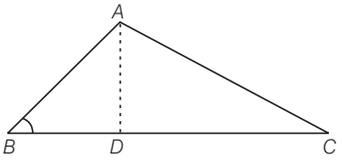
$$\angle ABC + \angle ACD = 2\angle AEC.$$

5. In a ΔABC, if side BC is produced to D and bisectors of $\angle ABC$ and $\angle ACD$ meet at E, then



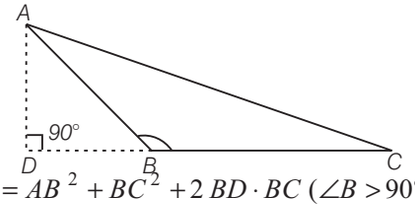
$$\angle BEC = \frac{1}{2}\angle BAC$$

6. In an acute angle ΔABC, AD is a perpendicular dropped on the opposite side of $\angle A$, then



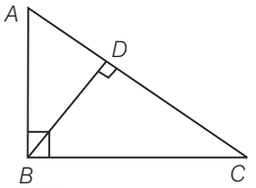
$$AC^2 = AB^2 + BC^2 - 2BD \cdot BC \ (\angle B < 90°)$$

7. In an obtuse angle ΔABC, AD is perpendicular dropped on BC. BC is produced to D to meet AD, then



$$AC^2 = AB^2 + BC^2 + 2\,BD \cdot BC \;(\angle B > 90°)$$

8. In a right angle ΔABC, $\angle B = 90°$ and AC is hypotenuse. The perpendicular BD is dropped on hypotenuse AC from right angle vertex B, then



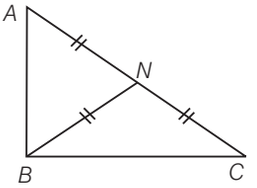
(i) $BD = \dfrac{AB \times BC}{AC}$ (ii) $AD = \dfrac{AB^2}{AC}$

(iii) $CD = \dfrac{BC^2}{AC}$

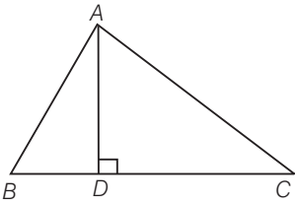
(iv) $\dfrac{1}{BD^2} = \dfrac{1}{AB^2} + \dfrac{1}{BC^2}$

In a right angled triangle, the median to the hypotenuse $= \dfrac{1}{2} \times$ hypotenuse

i.e., $BN = \dfrac{AC}{2}$ (as per the fig.)



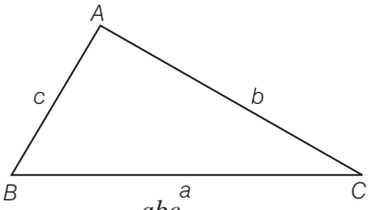
9. **Area of a triangle (General formula)**



$$A\,(\Delta) = \frac{1}{2} \times \text{base} \times \text{height}$$

$$A\,(\Delta) = \frac{1}{2} \times BC \times AD \qquad \text{(as per the figure.)}$$

10. **Area of scalene triangle** $= \sqrt{s\,(s-a)\,(s-b)\,(s-c)}$



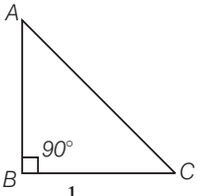
Also, $A\,(\Delta) = r \times s = \dfrac{abc}{4R}$

where a, b and c are the sides of the triangle.

$s \rightarrow$ semiperimeter $= \dfrac{a+b+c}{2}$, $r \rightarrow$ inradius

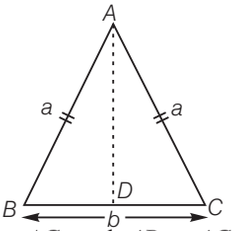
$R \rightarrow$ circumradius

11. **Area of right angled triangle**



$= \dfrac{1}{2} \times$ base $\times$ height $= \dfrac{1}{2} \times BC \times AB$ (as per the figure)

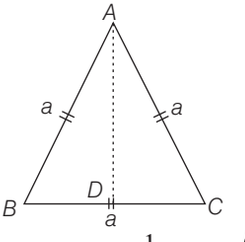
12. **Area of an isosceles triangle** $= \dfrac{b}{4}\sqrt{4a^2 - b^2}$



$AB = AC$ and $\angle B = \angle C$

$\Delta ABD \cong \Delta ACD$ ($AD \rightarrow$ Angle bisector, median, altitude and perpendicular bisector)

13. **Area of an equilateral triangle** $= \dfrac{\sqrt{3}}{4}a^2$



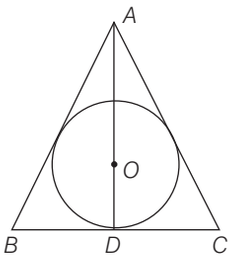
$$\left[A\,(\Delta) = \frac{1}{2}BC \times AD = \frac{1}{2} \times a \times \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{4}a^2\right]$$

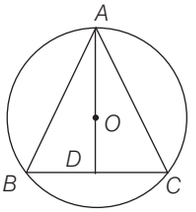
($a \rightarrow$ each side of the triangle)

$AD \rightarrow$ Altitude, median, angle bisector and perpendicular bisector also.

Inradius : $\frac{1}{3} \times \text{height} = \frac{\text{side}}{2\sqrt{3}}$,

$OD \rightarrow$ Inradius





Circumradius $= \frac{2}{3} \times \text{height} = \frac{\text{side}}{\sqrt{3}}$

$OA \rightarrow$ Circumradius

NOTE In equilateral triangle orthocentre centroid, incentre and circumcentre coincide at the same point.

Circumradius = 2 × inradius

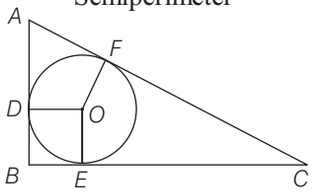
For the given perimeter of a triangle, the area of equilateral triangle is maximum.

For the given area of a triangle, the perimeter of equilateral triangle is minimum.

14. In a right angled triangle

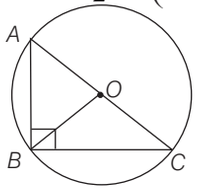
(i) Inradius $(r) = \frac{AB + BC - AC}{2}$

(ii) Inradius $(r) = \frac{\text{Area}}{\text{Semiperimeter}}$



$DO = EO = FO\ (r)$

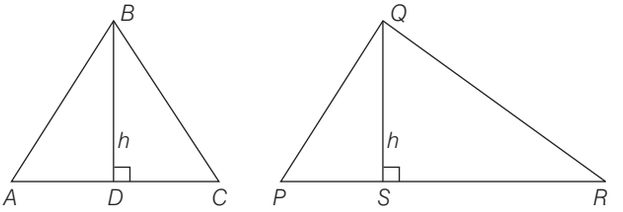
(iii) Circumradius $(R) = \frac{AC}{2} = \left(\frac{\text{hypotenuse}}{2}\right)$



$AO = CO = BO = (R)$

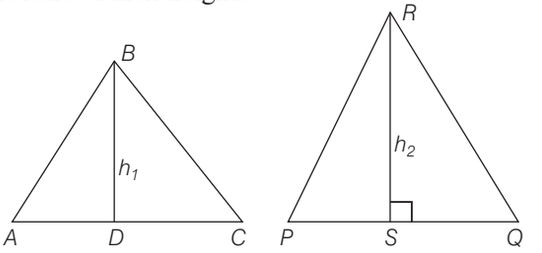
AC is the diameter.

15. The ratio of areas of two triangles of equal heights is equal to the ratio of their corresponding bases. i.e.,



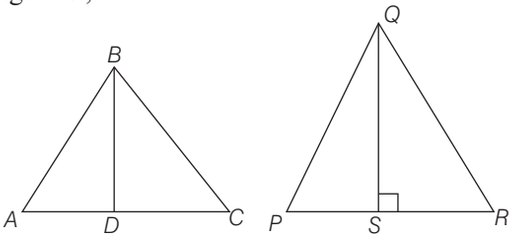
$$\frac{A\,(\Delta ABC)}{A\,(\Delta PQR)} = \frac{AC}{PR}$$

16. The ratio of areas of triangles of equal bases is equal to the ratio of their heights.



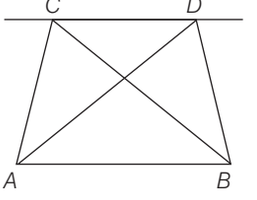
i.e., $\frac{A\,(\Delta ABC)}{A\,(\Delta PQR)} = \frac{BD}{RS}$

17. The ratio of the areas of two triangles is equal to the ratio of the products of base and its corresponding height i.e.,



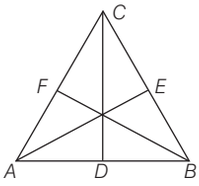
$$\frac{A\,(\Delta ABC)}{A\,(\Delta PQR)} = \frac{AC \times BD}{PR \times QS}$$

18. If the two triangles have the same base and lie between the same parallel lines (as shown in figure), then the area of two triangles will be equal.



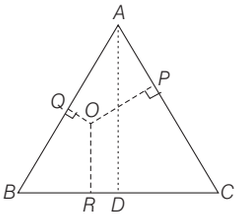
i.e., $A\,(\Delta ABC) = A\,(\Delta ADB)$

19. In a triangle AE, CD and BF are the medians, then

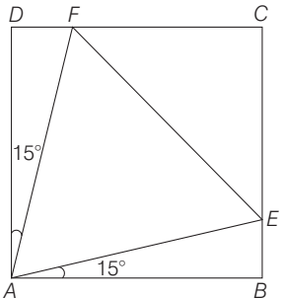


$$3\,(AB^2 + BC^2 + AC^2) = 4\,(CD^2 + BF^2 + AE^2)$$

20. In an equilateral triangle, if O is a point anywhere inside the equilateral triangle ABC, the sum of its distances from three sides is equal to the length of the altitude of the triangle. That is, as per the given diagram, $OP + OQ + OR = AD$.



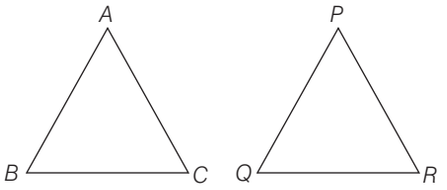
21. The largest possible area of an equilateral triangle inscribed in a unit square is $(2\sqrt{3} - 3)$.



- The maximum area can be enclosed only when one of the vertices of the triangle coincides with one of the vertices of the square and angle between the side of the triangle and the side of the rectangle at the point of coincidence is 15°.
- Also, the side of the equilateral triangle is $\sqrt{6} - \sqrt{2}$.
- Also, the area of $\Delta ADF = \Delta ABE = \frac{1}{2}(\Delta ECF)$.

Congruency of triangles

Two triangles are said to be congruent if they are equal in all respects. i.e.,

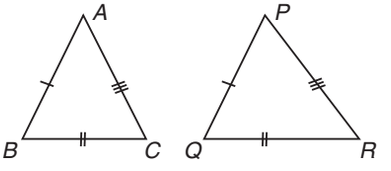
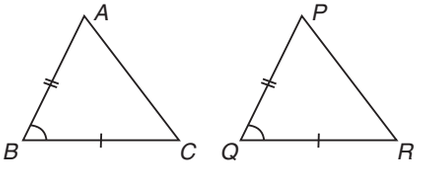


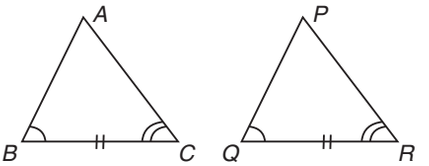
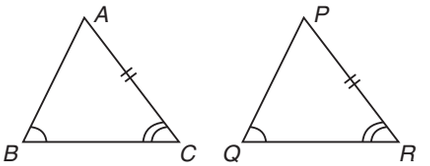
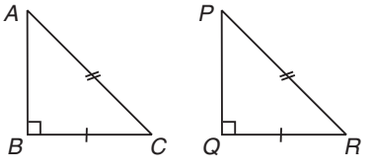
1. Each of the three sides of one triangle must be equal to the three respective sides of the other.
2. Each of the three angles of the one triangle must be equal to the three respective angles of the other.

$$\text{i.e.,} \quad \left.\begin{matrix} AB = PQ \\ AC = PR \\ BC = QR \end{matrix}\right\} \quad \text{and} \quad \left.\begin{matrix} \angle A = \angle P \\ \angle B = \angle Q \\ \angle C = \angle R \end{matrix}\right\}$$

Tests for congruency

With the help of the following given tests, we can deduce without having detailed information about triangles that whether the given two triangles are congruent or not.

Test	Property	Diagram
$S - S - S$	(Side–Side–Side) If the three sides of one triangle are equal to the corresponding three sides of the other triangle, then the two triangles are congruent. $AB \cong PQ,\ AC \cong PR,\ BC \cong QR$ $\therefore \quad \Delta ABC \cong \Delta PQR$	
$S - A - S$	(Side–Angle–Side) If two sides and the angle included between them are congruent to the corresponding sides and the angle included between them, of the other triangle, then the two triangles are congruent. $AB \cong PQ,\ \angle ABC \cong \angle PQR,\ BC \cong QR$ $\therefore \quad \Delta ABC \cong \Delta PQR$	

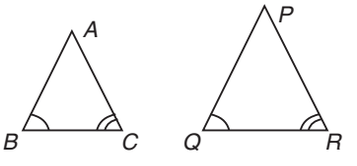
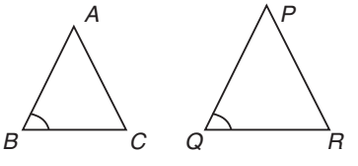
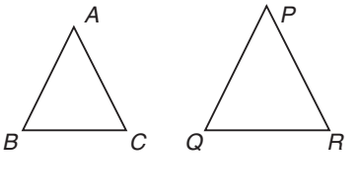
Test	Property	Diagram
$A-S-A$	(Angle–Side–Angle) If two angles and the included side of a triangle are congruent to the corresponding angles and the included side of the other triangle, then the two triangles are congruent. $\angle ABC \cong \angle PQR,\ BC \cong QR,\ \angle ACB \cong \angle PRQ$ $\therefore \quad \Delta ABC \cong \Delta PQR$	
$A-A-S$	(Angle–Angle–Side) If two angles and a side other than the included side of a one triangle are congruent to the corresponding angles and a corresponding side other than the included side of the other triangle, then the two triangles are congruent. $\angle ABC \cong \angle PQR,\ \angle ACB \cong \angle PRQ$ and $AC \cong PR$ (or $AB \cong PQ$)	
$R—H—S$	(Right angle–Hypotenuse–Side) If the hypotenuse and one side of the right angled triangle are congruent to the hypotenuse and a corresponding side of the other right angled triangle, then the two given triangles are congruent. $AC \cong PR,\ \angle B = \angle Q$ and $BC \cong QR$ $\therefore \quad \Delta ABC \cong \Delta PQR$	

Similarity of triangles

Two triangles are said to be similar if the corresponding **angles** are **congruent** and their corresponding **sides** are in **proportion**. The symbol for similarity is '~'. If $\Delta ABC \sim \Delta PQR$, then $\angle ABC \cong \angle PQR, \quad \angle BCA \cong \angle QRP, \quad \angle BAC \cong \angle QPR$

Tests for Similarity

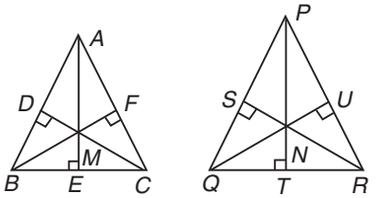
Through the tests for similarity we can deduce the similarity of triangles with minimum required information.

Test	Property/Definition	Diagram
$A-A$	Angle-Angle If the two angles of one triangle are congruent to the corresponding two angles of the other triangle, then the two triangles are said to be similar. $\angle ABC \cong \angle PQR$ $\angle ACB \cong \angle PRQ$ $\therefore \quad \Delta ABC \sim \Delta PQR$	
$S\text{-}A\text{-}S$	Side-Angle-Side If the two sides of one triangle are proportional to the corresponding two sides of the other triangle and the angle included by them are congruent, then the two triangles are similar. *i.e.,* $\frac{AB}{PQ} = \frac{BC}{QR}$ and $\angle ABC = \angle PQR$ $\therefore \quad \Delta ABC \sim \Delta PQR.$	 $\frac{AB}{PQ} = \frac{BC}{QR} = K$ (K is any constant)
$S\text{-}S\text{-}S$	Side-Side-Side If the three sides of one triangle are proportional to the corresponding three sides of the other triangle, then the two triangles are similar. *i.e.,* $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$ $\therefore \quad \Delta ABC \sim \Delta PQR$	 $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = K$

NOTE When the corresponding sides are in proportion , then the corresponding angles are in proportion.

Properties of Similar Triangles

If the two triangles are similar, then for the proportional/corresponding sides we have the following results.

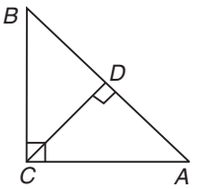


1. Ratio of sides = Ratio of heights (altitudes)
   = Ratio of medians
   = Ratio of angle bisectors
   = Ratio of inradii
   = Ratio of circumradii

2. Ratio of areas = Ratio of squares of corresponding sides.
   i.e., if $\Delta ABC \sim \Delta PQR$, then
   $$\frac{A(\Delta ABC)}{A(\Delta PQR)} = \frac{(AB)^2}{(PQ)^2} = \frac{(BC)^2}{(QR)^2} = \frac{(AC)^2}{(PR)^2}$$

NOTE Rule 1 can also apply with rule 2.

3. In a right angled triangle, the triangles on each side of the altitude drawn from the vertex of the right angle to the hypotenuse are similar to the original triangle and to each other too. *i.e.,* $\Delta BCA \sim \Delta BDC \sim \Delta CDA$.

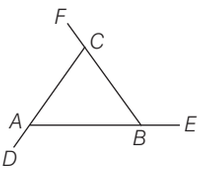


Introductory *Exercise 12.2*

1. In a triangle ABC, if AB, BC and AC are the three sides of the triangle, then which of the statements is necessarily true?
(a) $AB + BC < AC$ (b) $AB + BC > AC$
(c) $AB + BC = AC$ (d) $AB^2 + BC^2 = AC^2$

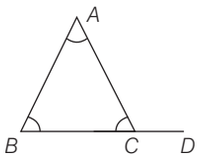
2. The sides of a triangle are 12 cm, 8 cm and 6 cm respectively, the triangle is :
(a) acute (b) obtuse
(c) right (d) can't be determined

3. If the sides of a triangle are produced then the sum of the exterior angles *i.e,*
$\angle DAB + \angle EBC + \angle FCA$ is equal to :



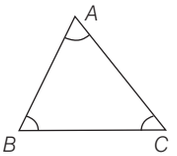
(a) 180° (b) 270° (c) 360° (d) 240°

4. In the given figure BC is produced to D and $\angle BAC = 40°$ and $\angle ABC = 70°$. Find the value of $\angle ACD$.



(a) 30° (b) 40°
(c) 70° (d) 110°

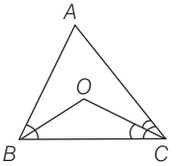
5. In a ΔABC, $\angle BAC > 90°$, then $\angle ABC$ and $\angle ACB$ must be :



(a) acute
(b) obtuse
(c) one acute and one obtuse
(d) can't be determined

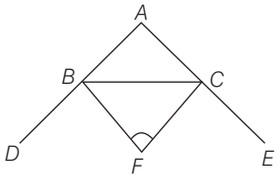
6. If the angles of a triangle are in the ratio 1 : 4 : 7, then the value of the largest angle is :
(a) 135°
(b) 84°
(c) 105°
(d) none of the above

7. In the adjoining figure $\angle B = 70°$ and $\angle C = 30°$. BO and CO are the angle bisectors of $\angle ABC$ and $\angle ACB$. Find the value of $\angle BOC$.



(a) 30° (b) 40°
(c) 120° (d) 130°

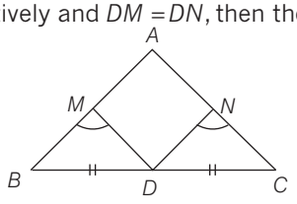
8. In the given diagram of ΔABC, $\angle B = 80°$, $\angle C = 30°$. BF and CF are the angle bisectors of $\angle CBD$ and $\angle BCE$ respectively. Find the value of $\angle BFC$.



(a) 110° (b) 50° (c) 125° (d) 55°

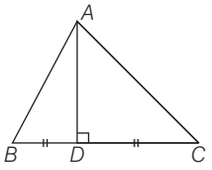
9. In an equilateral triangle, the incentre, circumcentre, orthocentre and centroid are :
(a) concylic (b) coincident
(c) collinear (d) None of these

10. In the adjoining figure, D is the midpoint of BC of a ΔABC. DM and DN are the perpendiculars on AB and AC respectively and $DM = DN$, then the ΔABC is :



(a) right angled (b) isosceles
(c) equilateral (d) scalene

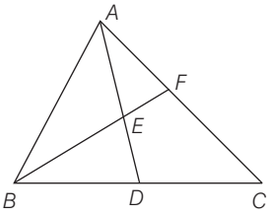
11. In the adjoining figure of ΔABC, AD is the perpendicular bisector of side BC. The triangle ABC is:
(a) right angled
(b) isosceles
(c) scalene
(d) equilateral



12. Triangle ABC is such that $AB = 9$ cm, $BC = 6$ cm, $AC = 7.5$ cm. Triangle DEF is similar to ΔABC. If $EF = 12$ cm then DE is :
(a) 6 cm (b) 16 cm (c) 18 cm (d) 15 cm

13. In ΔABC, $AB = 5$ cm, $AC = 7$ cm. If AD is the angle bisector of $\angle A$. Then $BD : CD$ is :
(a) 25 : 49 (b) 49 : 25 (c) 6 : 1 (d) 5 : 7

14. In a ΔABC, D is the mid-point of BC and E is mid-point of AD, BF passes through E. What is the ratio of $AF : FC$?



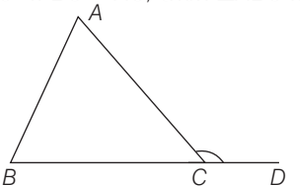
(a) 1 : 1 (b) 1 : 2 (c) 1 : 3 (d) 2 : 3

15. In a ΔABC, $AB = AC$ and $AD \perp BC$, then :
(a) $AB < AD$ (b) $AB > AD$
(c) $AB = AD$ (d) $AB \leq AD$

16. The difference between altitude and base of a right angled triangle is 17 cm and its hypotenuse is 25 cm. What is the sum of the base and altitude of the triangle is :
(a) 24 cm (b) 31 cm
(c) 34 cm (d) can't be determined

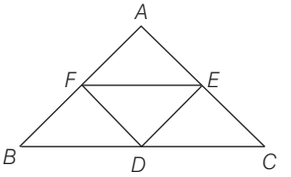
17. If AB, BC and AC be the three sides of a triangle ABC, then which one of the following is true?
(a) $AB - BC = AC$ (b) $(AB - BC) > AC$
(c) $(AB - BA) < AC$ (d) $AB^2 - BC^2 = AC^2$

18. In the triangle ABC, side BC is produced to D. $\angle ACD = 100°$ if $BC = AC$, then $\angle ABC$ is :



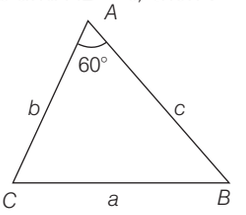
(a) 40°
(b) 50°
(c) 80°
(d) can't be determined

19. In the adjoining figure D, E and F are the mid-points of the sides BC, AC and AB respectively. ΔDEF is congruent to triangle :



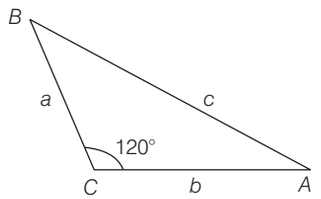
(a) ABC (b) AEF
(c) CDE, BFD (d) AFE, BFD and CDE

20. In the adjoining figure, $\angle BAC = 60°$ and $BC = a$, $AC = b$ and $AB = c$, then :



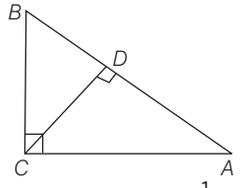
(a) $a^2 = b^2 + c^2$
(b) $a^2 = b^2 + c^2 - bc$
(c) $a^2 = b^2 + c^2 + bc$
(d) $a^2 = b^2 + 2bc$

21. In the adjoining figure of ΔABC, $\angle BCA = 120°$ and $AB = c$, $BC = a$, $AC = b$, then:



(a) $c^2 = a^2 + b^2 + ba$
(b) $c^2 = a^2 + b^2 - ba$
(c) $c^2 = a^2 + b^2 - 2ba$
(d) $c^2 = a^2 + b^2 + 2ab$

22. In a right angled ΔABC, $\angle C = 90°$ and CD is the perpendicular on the hypotenuse AB, $AB = c$, $BC = a$, $AC = b$ and $CD = p$, then :



(a) $\frac{p}{a} = \frac{p}{b}$ (b) $\frac{1}{p^2} + \frac{1}{b^2} = \frac{1}{a^2}$
(c) $p^2 = b^2 + c^2$ (d) $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

23. If the medians of a triangle are equal, then the triangle is :
(a) right angled (b) isoscles
(c) equilateral (d) scalene

24. The incentre of a triangle is determined by the :
(a) medians
(b) angle bisector
(c) perpendicular bisectors
(d) altitudes

25. The circumcentre of a triangle is determined by the :
(a) altitudes
(b) median
(c) perpendicular bisector
(d) angle bisectors

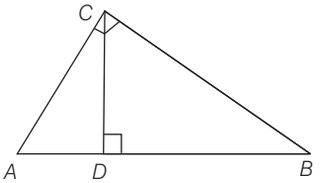
26. The point of intersection of the angle bisectors of a triangle is :
(a) orthocentre (b) centroid
(c) incentre (d) circumcentre

27. A triangle PQR is formed by joining the mid-points of the sides of a triangle ABC. 'O' is the circumcentre of ΔABC, then for ΔPQR, the point 'O' is :
(a) incentre (b) circumcentre
(c) orthocentre (d) centroid

28. If in a ΔABC, 'S' is the circumcentre, then :
(a) S is equidistant from all the vertices of a triangle
(b) S is equidistant from all the sides of a triangle
(c) AS, BS and CS are the angular bisectors
(d) AS, BS and CS produced are the altitudes on the opposite sides.

29. If AD, BE, CF are the altitudes of ΔABC whose orthocentre is H, then C is the orthocentre of :
(a) ΔABH (b) ΔBDH
(c) ΔABD (d) ΔBEA

30. In a right angled ΔABC, $\angle C = 90°$ and CD is the perpendicular on hypotenuse AB if $BC = 15$ cm and $AC = 20$ cm, then CD is equal to :



(a) 18 cm (b) 12 cm
(c) 17.5 cm (d) can't be determined

31. In an equilateral ΔABC, if a, b and c denote the lengths of perpendiculars from A, B and C respectively on the opposite sides, then :
(a) $a > b > c$ (b) $a > b < c$
(c) $a = b = c$ (d) $a = c \neq b$

32. What is the ratio of side and height of an equilateral triangle?
(a) 2 : 1 (b) 1 : 1
(c) $2 : \sqrt{3}$ (d) $\sqrt{3} : 2$

33. The triangle is formed by joining the mid-points of the sides AB, BC and CA of ΔABC and the area of ΔPQR is 6 cm^2, then the area of ΔABC is :
(a) 36 cm^2 (b) 12 cm^2
(c) 18 cm^2 (d) 24 cm^2

34. One side other than the hypotenuse of right angle isosceles triangle is 6 cm. The length of the perpendicular on the hypotenuse from the opposite vertex is :
(a) 6 cm (b) $6\sqrt{2}$ cm
(c) 4 cm (d) $3\sqrt{2}$ cm

35. Any two of the four triangles formed by joining the mid- points of the sides of a given triangle are :
(a) congruent
(b) equal in area but not congruent
(c) unequal in area and not congruent
(d) None of the above

36. The internal bisectors of $\angle B$ and $\angle C$ of ΔABC meet at O. If $\angle A = 80°$ then $\angle BOC$ is :
(a) 50° (b) 160°
(c) 100° (d) 130°

37. The point in the plane of a triangle which is at equal perpendicular distance from the sides of the triangle is :
(a) centroid (b) incentre
(c) circumcenre (d) orthocentre

38. Incentre of a triangle lies in the interior of :
(a) an isosceles triangle only
(b) a right angled triangle only
(c) any equilateral triangle only
(d) any triangle

39. In a triangle PQR, PQ=20 cm and PR = 6 cm, the side QR is :
(a) equal to 14 cm
(b) less than 14 cm
(c) greater than 14 cm
(d) none of the above

40. The four triangles formed by joining the pairs of mid-points of the sides of a given triangle are congruent if the given triangle is :
(a) an isosceles triangle
(b) an equilateral triangle
(c) a right angled triangle
(d) of any shape

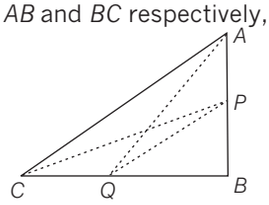
41. O is orthocentre of a triangle PQR, which is formed by joining the mid-points of the sides of a ΔABC, O is :
(a) orthocentre (b) incentre
(c) circumcentre (d) centroid

42. In a ΔABC, a line PQ parallel to BC cuts AB at P and AC at Q. If BQ bisects $\angle PQC$, then which one of the following relation is always true :
(a) $BC = CQ$ (b) $BC = BQ$
(c) $BC \neq CQ$ (d) $BC \neq BQ$

43. Which of the following is true, in the given figure, where AD is the altitude to the hypotenuse of a right angled ΔABC?
(i) ΔCAD and ΔABD are similar
(ii) ΔCDA and ΔADB are congruent
(iii) ΔADB and ΔCAB are similar
Select the correct answer using the codes given below :
(a) 1 and 2
(b) 2 and 3
(c) 1 and 3
(d) 1, 2 and 3

44. If D is such a point on the side BC of ΔABC that $\frac{AB}{AC} = \frac{BD}{CD}$, then AD must be a/an :
(a) altitude of ΔABC
(b) median of ΔABC
(c) angle bisector of ΔABC
(d) perpendicular bisector of ΔABC

45. In right angled ΔABC, $\angle B = 90°$, if P and Q are points on the sides AB and BC respectively, then :



(a) $AQ^2 + CP^2 = 2\,(AC^2 + PQ^2)$
(b) $AQ^2 + CP^2 = AC^2 + PQ^2$
(c) $(AQ^2 + CP^2) = \frac{1}{2}\,(AC^2 + PQ^2)$
(d) $(AQ + CP) = \frac{1}{2}\,(AC + PQ)$

46. If ABC is a right angled triangle at B and M, N are the mid-points of AB and BC, then 4 $(AN^2 + CM^2)$ is equal to :
(a) $4AC^2$ (b) $6AC^2$
(c) $5AC^2$ (d) $\frac{5}{4}AC^2$

47. If ΔABC and ΔDEF are so related that $\frac{AB}{FD} = \frac{BC}{DE} = \frac{CA}{EF}$, then which of the following is true ?
(a) $\angle A = \angle F$ and $\angle B = \angle D$
(b) $\angle C = \angle F$ and $\angle A = \angle D$
(c) $\angle B = \angle F$ and $\angle C = \angle D$
(d) $\angle A = \angle E$ and $\angle B = \angle D$

48. ABC is a right angle triangle at A and AD is perpendicular to the hypotenuse. Then $\frac{BD}{CD}$ is equal to :
(a) $\left(\frac{AB}{AC}\right)^2$ (b) $\left(\frac{AB}{AD}\right)^2$ (c) $\frac{AB}{AC}$ (d) $\frac{AB}{AD}$

49. Let ABC be an equilateral triangle. Let $BE \perp CA$ meeting CA at E, then $(AB^2 + BC^2 + CA^2)$ is equal to :
(a) $2BE^2$ (b) $3BE^2$ (c) $4BE^2$ (d) $6BE^2$

50. If D, E and F are respectively the mid-points of sides BC, CA and AB of a ΔABC. If EF = 3 cm, FD = 4 cm and AB = 10 cm, then DE, BC and CA respectively will be equal to :
(a) 6, 8 and 20 cm (b) 4, 6 and 8 cm
(c) 5, 6 and 8 cm (d) $\frac{10}{3}$, 9 and 12 cm

51. In the right angle triangle $\angle C = 90°$. AE and BD are two medians of a triangle ABC meeting at F. The ratio of the area of ΔABF and the quadrilateral $FDCE$ is :
(a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 2 : 3

52. ABC is a triangle and DE is drawn parallel to BC cutting the other sides at D and E. If $AB = 3.6$ cm, $AC = 2.4$ cm and $AD = 2.1$ cm, then AE is equal to :

(a) 1.4 cm (b) 1.8 cm
(c) 1.2 cm (d) 1.05 m

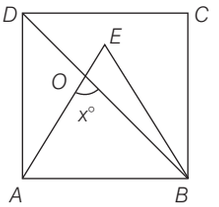
53. Consider the following statements :

(1) If three sides of a triangle are equal to three sides of another triangle, then the triangles are congruent.

(2) If three angles of a triangle are equal to three angles of another triangle respectively, then the two triangles are congruent. Out of these statements :

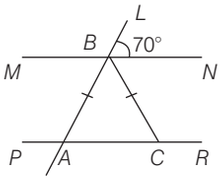
(a) 1 is correct and 2 is false
(b) both 1and 2 are false
(c) both 1 and 2 are correct
(d) 1 is false and 2 is correct

54. In the figure ΔABE is an equilateral triangle in a square $ABCD$. Find the value of angle x in degrees :



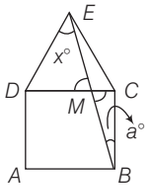
(a) 60° (b) 45° (c) 75° (d) 90°

55. In the given diagram $MN \parallel PR$ and $m\angle LBN = 70°$, $AB = BC$. Find $m\angle ABC$:



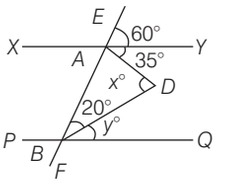
(a) 40° (b) 30°
(c) 35° (d) 55°

56. In the given diagram, equilateral triangle EDC surmounts square $ABCD$. Find the $m\angle BED$ represented by x where $m\angle EBC = \alpha°$.



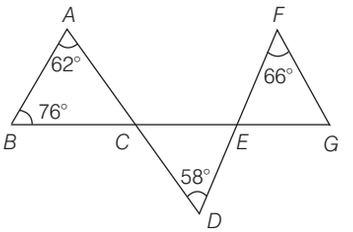
(a) 45° (b) 60°
(c) 30° (d) none of these

57. In the given diagram $XY \parallel PQ$. Find $m\angle x°$ and $m\angle y°$.



(a) 75° and 40° (b) 45°, 60°
(c) 75°, 45° (d) 60° and 45°

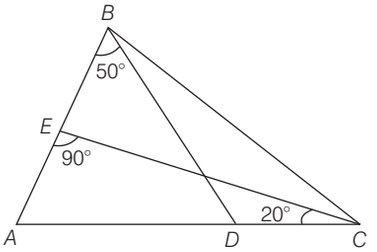
58. In the adjoining figure $m\angle CAB = 62°$, $m\angle CBA = 76°$ $m\angle ADE = 58°$ and $\angle DFG = 66°$,



Find $m\angle FGE$.

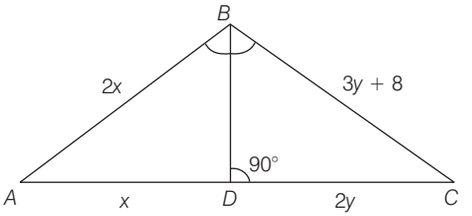
(a) 44° (b) 34°
(c) 36° (d) none of these

59. In the given figure $CE \perp AB$, $m\angle ACE = 20°$ and $m\angle ABD = 50°$. Find $m\angle BDA$.



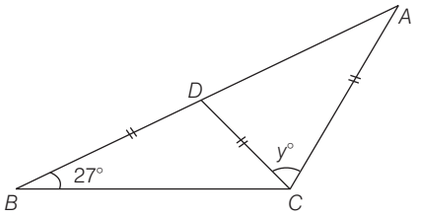
(a) 50° (b) 60°
(c) 70° (d) 80°

60. In the ΔABC, BD bisects $\angle B$, and is perpendicular to AC. If the lengths of the sides of the triangle are expressed in terms of x and y as shown,find the value of x and y.



(a) 6, 12 (b) 10, 12
(c) 16, 8 (d) 8, 15

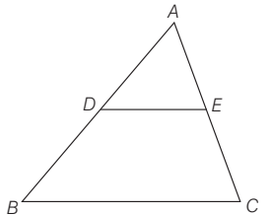
61. In the following figure $ADBC$. $BD = CD = AC$, $m\angle ABC = 27°$, $m\angle ACD = y$. Find the value of y.



(a) 27° (b) 54°
(c) 72° (d) 58°

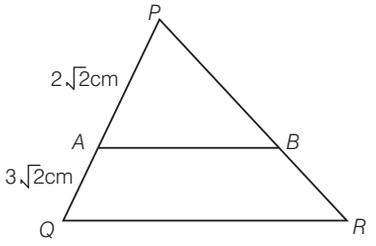
62. ABC is an isosceles triangle with $AB = AC$. Side BA is produced to D such that $AB = AD$. Find $m\angle BCD$.
(a) 60° (b) 90°
(c) 120° (d) can't be determined

63. In ΔABC, $AC = 5$ cm. Calculate the length of AE where $DE \parallel BC$. Given that $AD = 3$ cm and $BD = 7$ cm.



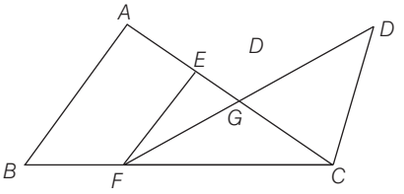
(a) 2 cm (b) 1 cm (c) 1.5 cm (d) 2.5 cm

64. In ΔPQR, $AP = 2\sqrt{2}$ cm, $AQ = 3\sqrt{2}$ cm and $PR = 10$ cm, $AB \parallel QR$. Find the length of BR.



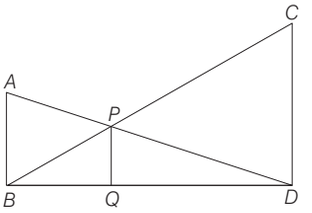
(a) $6\sqrt{2}$ cm (b) 6 cm
(c) $5\sqrt{2}$ cm (d) none of these

65. In the adjoining figure (not drawn to scale) AB, EF and CD are parallel lines. Given that $EG = 5$ cm, $GC = 10$ cm and $DC = 18$ cm. Calculate AC, if $AB = 15$ cm.



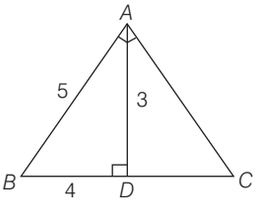
(a) 21 cm (b) 25 cm
(c) 18 cm (d) 28 cm

66. In the given figure, AB, PQ and CD are perpendiculars on the line segment BD. If $AB = x$, $CD = y$ and $BD = z$, find the length of PQ.



(a) $\frac{3xyz}{x+y+z}$ (b) $\frac{x+y}{z}$ (c) $\frac{xy}{x+y}$ (d) $\frac{x+y+z}{xyz}$

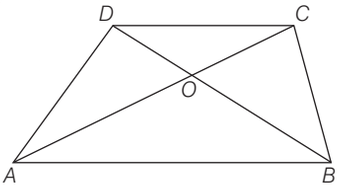
67. In the adjoining figure the angle BAC and $\angle ADB$ are right angles. $BA = 5$ cm, $AD = 3$ cm and $BD = 4$ cm, what is the length of DC ?



(a) 2.5 (b) 3 (c) 2.25 (d) 2

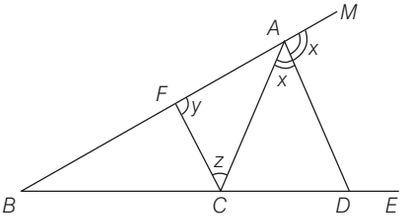
68. The areas of the similar triangles are in the ratio of 25 : 36. What is the ratio of their respective heights?
(a) 5 : 6 (b) 6 : 5 (c) 1 : 11 (d) 2 : 3

69. In the given diagram $AB \parallel CD$, then which one of the following is true ?



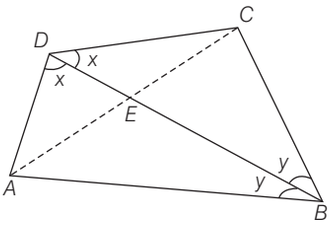
(a) $\frac{AB}{CD} = \frac{AO}{OC}$ (b) $\frac{AB}{CD} = \frac{BO}{OD}$
(c) $\Delta AOB \sim \Delta COD$ (d) All of these

70. The bisector of the exterior $\angle A$ of ΔABC intersects the side BC produced to D. Here CF is parallel to AD. Which one of the following is correct?



(a) $\frac{AB}{AC} = \frac{BD}{CD}$ (b) $\frac{AB}{AC} = \frac{CD}{BD}$
(c) $\frac{AB}{AC} = \frac{BC}{CD}$ (d) none of these

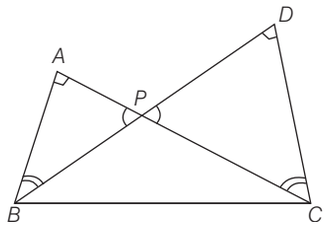
71. The diagonal BD of a quadrilateral $ABCD$ bisects $\angle B$ and $\angle D$, then :



(a) $\frac{AB}{CD} = \frac{AD}{BC}$ (b) $\frac{AB}{BC} = \frac{AD}{CD}$

(c) $AB = AD \times BC$ (d) none of these is true

72. Two right triangles ABC and DBC are drawn on the same hypotenuse BC on the same side of BC. If AC and DB intersects at P, then :



(a) $\frac{AP}{PC} = \frac{BP}{DP}$ (b) $AP \times DP = PC \times BP$

(c) $AP \times PC = BP \times DP$ (d) $AP \times BP = PC \times PD$

73. A man goes 150 m due east and then 200 m due north. How far is he from the starting point?

(a) 200 m (b) 350 m

(c) 250 m (d) 175 m

74. From a point O in the interior of a ΔABC perpendiculars OD, OE and OF are drawn to the sides BC, CA and AB respectively, then which one of the following is true ?

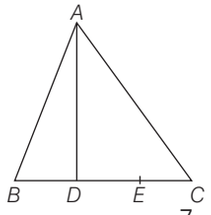
(a) $AF^2 + BD^2 + CE^2 = AE^2 + CD^2 + BF^2$

(b) $AB^2 + BC^2 = AC^2$

(c) $AF^2 + BD^2 + CE^2 = OA^2 + OB^2 + OC^2$

(d) $AF^2 + BD^2 + CE^2 = OD^2 + OE^2 + OF^2$

75. In an equilateral triangle ABC, the side BC is trisected at D. Find the value of AD^2 :



(a) $\frac{9}{7} AB^2$ (b) $\frac{7}{9} AB^2$

(c) $\frac{3}{4} AB^2$ (d) $\frac{4}{5} AB^2$

76. ABC is a triangle in which $\angle A = 90°$.

$AN \perp BC$, $AC = 12$ cm and $AB = 5$ cm. Find the ratio of the areas of ΔANC and ΔANB.

(a) 125 : 44 (b) 25 : 144

(c) 144 : 25 (d) 12 : 5

77. A vertical stick 15 cm long casts it shadow 10 cm long on the ground. At the same time a flag pole casts a shadow 60 cm long. Find the height of the flag pole.

(a) 40 cm (b) 45 cm

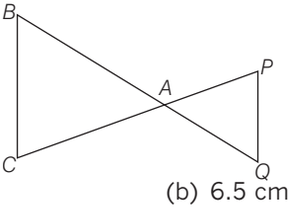
(c) 90 cm (d) none of these

78. Vertical angles of two isoceles triangles are equal. Then corresponding altitudes are in the ratio 4 : 9. Find the ratio of their areas :

(a) 16 : 49 (b) 16 : 81

(c) 16 : 65 (d) none of these

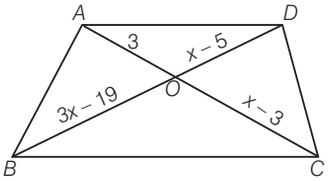
79. In the figure $\Delta ACB \sim \Delta APQ$. If $BC = 8$ cm, $PQ = 4$ cm, $AP = 2.8$ cm, find CA :



(a) 8 cm (b) 6.5 cm

(c) 5.6 cm (d) none of these

80. In the figure $BC \| AD$. Find the value of x.



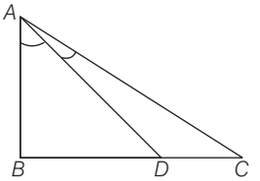
(a) 9, 10 (b) 7, 8 (c) 10, 12 (d) 8, 9

81. In an equilateral triangle of side $2a$, calculate the length of its altitude.

(a) $2a\sqrt{3}$ (b) $a\sqrt{3}$

(c) $a\frac{\sqrt{3}}{2}$ (d) none of these

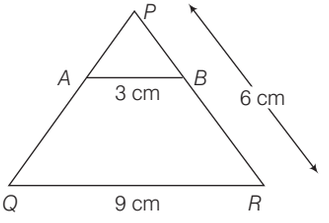
82. In figure AD is the bisector of $\angle BAC$. If $BD = 2$ cm, $CD = 3$ cm and $AB = 5$ cm. Find AC :



(a) 6 cm (b) 7.5 cm

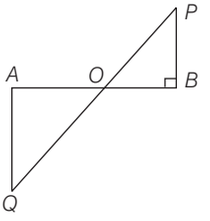
(c) 10 cm (d) 15 cm

83. In the figure $AB \parallel QR$. Find the length of PB.



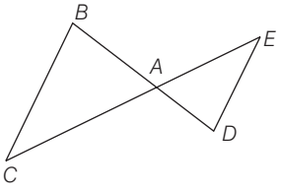
(a) 2 cm (b) 3 cm (c) 2.5 cm (d) 4 cm

84. In the figure QA and PB are perpendicular to AB. If $AO = 10$ cm, $BO = 6$ cm and $PB = 9$ cm. Find AQ.



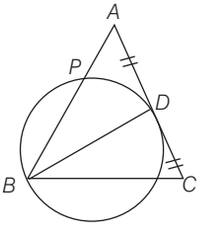
(a) 8 cm (b) 9 cm (c) 15 cm (d) 12 cm

85. In the given figure $AB = 12$ cm, $AC = 15$ cm and $AD = 6$ cm. $BC \parallel DE$, find the length of AE.



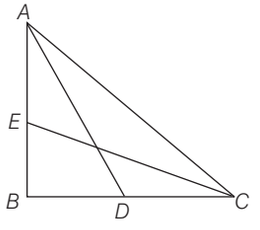
(a) 6 cm (b) 7.5 cm (c) 9 cm (d) 10 cm

86. In the figure, ABC is a triangle in which $AB = AC$. A circle through B touches AC at D and intersects AB at P. If D is the mid-point of AC, Find the value of AB :



(a) $2AP$
(b) $3AP$
(c) $4AP$
(d) none of the above

87. In figure, ABC is a right angled triangle, right angled at B. AD and CE are the two medians drawn from A and C respectively. If $AC = 5$ cm and $AD = 3\sqrt{5}/2$ cm, find the length of CE.



(a) $2\sqrt{5}$ cm (b) 2.5 cm (c) 5 cm (d) $4\sqrt{2}$ cm

88. In a ΔABC, $AB = 10$ cm, $BC = 12$ cm and $AC = 14$ cm. Find the length of median AD. If G is the centroid, find length of GA.

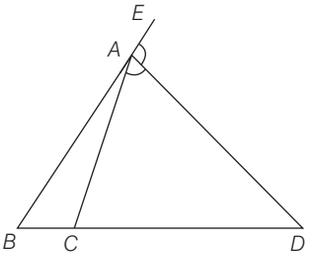
(a) $\frac{5}{3}\sqrt{7}, \frac{5}{9}\sqrt{7}$ (b) $5\sqrt{7}, 4\sqrt{7}$
(c) $\frac{10}{\sqrt{3}}, \frac{8}{3}\sqrt{7}$ (d) $4\sqrt{7}, \frac{8}{3}\sqrt{7}$

89. ΔABC is a right angled at A and AD is the altitude to BC. If $AB = 7$ cm and $AC = 24$ cm. Find the ratio of AD is to AM if M is the mid–point of BC.
(a) 25 : 41 (b) 32 : 41
(c) 336/625 (d) 625/336

90. Area of $\Delta ABC = 30$ cm^2. D and E are the mid-points of BC and AB respectively. Find $A(\Delta BDE)$.
(a) 10 cm^2 (b) 7.5 cm^2
(c) 15 cm^2 (d) none of these

91. The three sides of a triangle are given which one of the following is not a right angle?
(a) 20, 21, 29 (b) 16, 63, 65
(c) 56, 90, 106 (d) 36, 35, 74

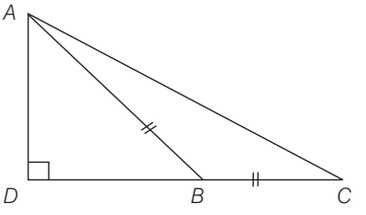
92. In the following figure, AD is the external bisector of $\angle EAC$, and it intersects BC produced to D. If $AB = 12$ cm, $AC = 8$ cm and $BC = 4$ cm, find CD.



(a) 10 cm (b) 6 cm
(c) 8 cm (d) 9 cm

93. In ΔABC, $AB^2 + AC^2 = 2500$ cm^2 and median $AD = 25$ cm, Find BC.
(a) 25 cm
(b) 40 cm
(c) 50 cm
(d) 48 cm

94. In the given figure, $AB = BC$ and $\angle BAC = 15°$, $AB = 10$ cm. Find the area of ΔABC.



(a) 50 cm^2 (b) 40 cm^2
(c) 25 cm^2 (d) 32 cm^2

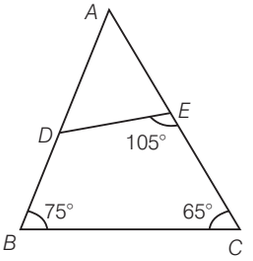
95. In ABC, G is the centroid, $AB = 15$ cm, $BC = 18$ cm and $AC = 25$ cm. Find GD, where D is the mid–point of BC.

(a) $\frac{1}{3}\sqrt{86}$ cm (b) $\frac{2}{3}\sqrt{86}$ cm

(c) $\frac{8}{3}\sqrt{15}$ cm (d) none of the above

96. In the given figure , if $\frac{DE}{BC} = \frac{2}{3}$ and if $AE = 10$ cm.

Find AB.



(a) 16 cm (b) 12 cm (c) 15 cm (d) 18 cm

97. Find the maximum area that can be enclosed in a triangle of perimeter 24 cm.

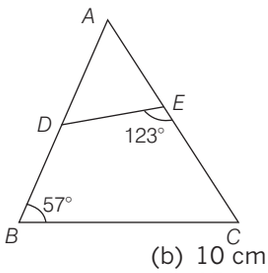
(a) 32 cm^2

(b) $16\sqrt{3}$ cm^2

(c) $16\sqrt{2}$ cm^2

(d) 27 cm^2

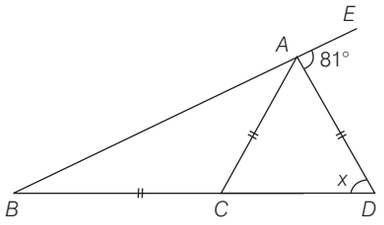
98. In the figure $AD = 12$ cm, $AB = 20$ cm and $AE = 10$ cm. Find EC.



(a) 14 cm (b) 10 cm

(c) 8 cm (d) 15 cm

99. In the given figure, $BC = AC = AD$, $\angle EAD = 81°$. Find the value of x.



(a) 45° (b) 54° (c) 63° (d) 36°

100. What is the ratio of inradius to the circumradius of a right angled triangle?

(a) 1 : 2 (b) $1 : \sqrt{2}$

(c) 2 : 5 (d) can't be determined

101. In a ΔABC, $AC = BC$. Point D lies on BC such that $CD = AD = AB$. Find $\angle ACB$.

(a) 18° (b) 30°

(c) 36° (d) 48°

102. In ΔABC, AD is the angle bisector of AB and a median to BC. If $AC = 15$ cm, find AB.

(a) 12 cm (b) 18 cm

(c) 15 cm (d) can't be determined

103. In a ΔABC, $\frac{(\angle BAC)}{(\angle ACB)} = \frac{1}{2}$ and $AC = 4$ cm, $BC = 6$ cm. Find AB.

(a) 6 (b) 10

(c) 9 (d) none of these

104. In a ΔABC, BD is the angle bisector of $\angle ABC$ and E is a point on AB such that $ED = AD$ and

$$\frac{(BD)\,(BC)}{CD} = \frac{(BE)\,(AB)}{AE}.$$

If $\angle A + \angle C = 66°$, what is the value of $\angle A - \angle C$?

(a) 10° (b) 22°

(c) 12° (d) 20°

105. Find the area of an isosceles triangle ABC in which $\angle ABC = 120°$ and each of the smaller side is 1 unit.

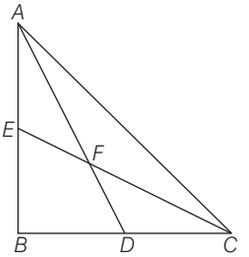
(a) $\frac{\sqrt{3}}{4}$ (b) $\frac{\sqrt{3}}{2}$ (c) $2\sqrt{3}$ (d) $\frac{3\sqrt{3}}{10}$

106. If O is the incentre of a triangle ABC and $\angle AOC = 124°$, find the $\angle ABC$.

(a) 56 (b) 112

(c) 68 (d) can't be determined

107. The following diagram depicts the right angle triangle ABC, where line segments AD and CE are the medians intersecting at a point F. Find the ratio of the area of ΔAEF to that of quadrilateral $BEFD$.



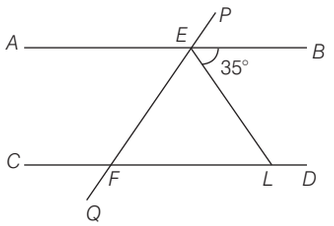
(a) 2 : 3 (b) 4 : 9

(c) 1 : 2 (d) none of these

41 $\angle BEF = 2\angle BEL = 70°$

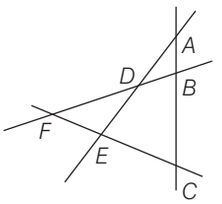
$\therefore \quad \angle CFQ = \angle EFL = 180° - 70°$

$= 110°$



44 ${}^4C_3 = \dfrac{4!}{3!1!} = \dfrac{4\cdot 3\cdot 2\cdot 1}{3\cdot 2\cdot 1\cdot 1} = 4$

45 ${}^4C_2 = \dfrac{4!}{2!2!} = \dfrac{4\cdot 3\cdot 2\cdot 1}{(2\cdot 1)\cdot(2\cdot 1)} = 6$



☑ Introductory Exercise 12.2

1 The sum of the smaller sides cannot be equal to or less than the largest side.

2 $12^2 > 8^2 + 6^2 \quad \Rightarrow$ obtuse triangle

3 Sum of exterior angles = 360°

4 $\angle ACD = \angle CBA + \angle BAC = 70° + 40° = 110°$

5 $\angle ABC + \angle ACB < 90°$

6 $x + 4x + 7x = 180°$

$\Rightarrow \quad x = 15°$

$\therefore \quad 7x = 105°$

7 $\angle OBC + \angle OCB = \dfrac{70}{2} + \dfrac{30}{2} = 50°$

$\therefore \quad \angle BOC = 180° - 50° = 130°$

8 $\angle BFC = 90° - \dfrac{1}{2}(\angle A)$

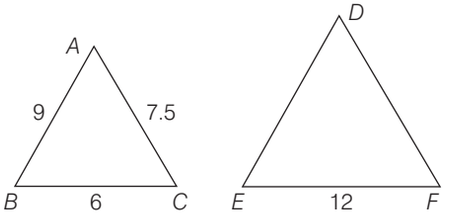
$= 90° - \dfrac{1}{2}[180° - (80° + 30°)]$

$= 55°$

10 Prove with the help of similarity.

11 Prove with the help of the similarity.

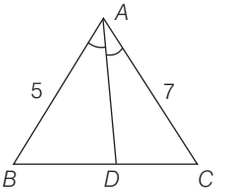
12 $\Delta ABC \sim \Delta DEF$



$\therefore \quad \dfrac{AB}{BC} = \dfrac{DE}{EF}$

$\Rightarrow \quad \dfrac{9}{6} = \dfrac{DE}{12} \Rightarrow DE = 18$ cm

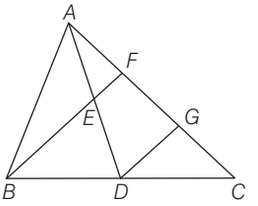
13 $\dfrac{AB}{AC} = \dfrac{BD}{CD} = \dfrac{5}{7}$ (By angle bisector theorem)



14 Let's draw a line segment DG, which is parallel to the line segment BF, such that G lies on AC.

Now look at the ΔADG, where $EF \parallel DG$, so we have

$$\frac{AF}{FG} = \frac{AE}{DE} = \frac{1}{1}$$



Now, let's have a look at the ΔBFC, where $BF \parallel DG$, so we have

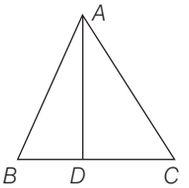
$$\frac{FG}{GC} = \frac{BD}{BC} = \frac{1}{1}$$

Thus we can say that $AF : FG : GC = 1 : 1 : 1$

Therefore, $AF : FC = 1 : 2$

15 AB is hypotenuse and AD is perpendicular in ΔABD,

$\therefore \quad AB > AD$

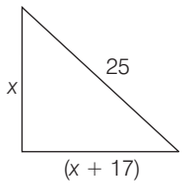


16 $x^2 + (x-17)^2 = 25^2$

(Using Pythagoras theorem)

$\therefore \quad x = 24$ cm

$\therefore$ Altitude + Base = 24 + 7 = 31 cm



17 Go back to the basics.

18 $\angle ACB = 180° - 100° = 80°$

and $\angle BAC + \angle ABC = 180° - 80° = 100°$

$\therefore \quad \angle ABC = 50° \quad (\because \angle BAC = \angle ABC)$

20 $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$ (cosine rule)

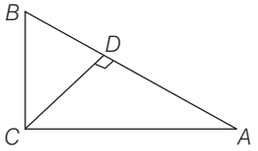
$\dfrac{1}{2} = \dfrac{b^2 + c^2 - a^2}{2bc} \quad (\because \cos 60° = 1/2)$

$\therefore \quad a^2 = b^2 + c^2 - bc$

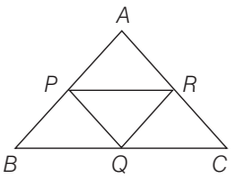
21 $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$

$-\dfrac{1}{2} = \dfrac{a^2 + b^2 - c^2}{2ab} \Rightarrow c^2 = a^2 + b^2 + ab$

22 $\dfrac{1}{p^2} = \dfrac{1}{a^2} + \dfrac{1}{b^2}$ (Apply similarity of triangles)

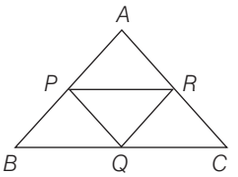


27



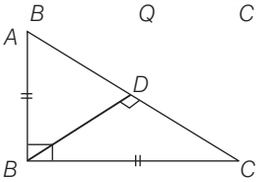
33 There are 4 congruent triangles.

Hence, $A(\Delta ABC) = 6 \times 4 = 24\ \text{cm}^2$



34 $BD = \dfrac{AB \times BC}{AC}$

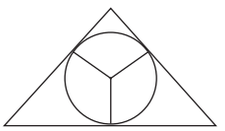
$= \dfrac{6 \times 6}{6\sqrt{2}} = 3\sqrt{2}$ cm



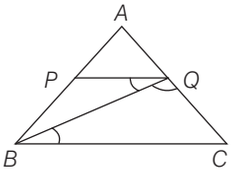
36 $\angle BOC = 90° + \dfrac{1}{2}\angle A$

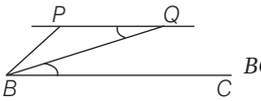
$= 90° + 40° = 130°$

37



42



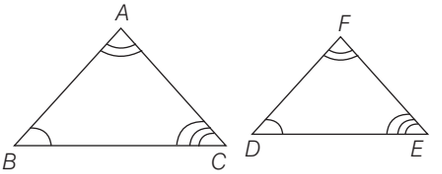


i.e., BQ is a transversal of PQ and BC.

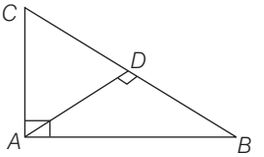
45 Apply Pythagoras theorem.

46 Apply Pythagoras theorem.

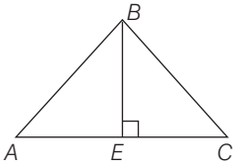
47



48 $\dfrac{BD}{CD} = \dfrac{(AB)^2}{(AC)^2}$



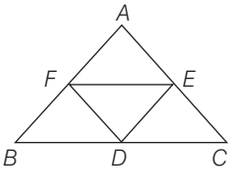
49 Apply Pythagoras theorem



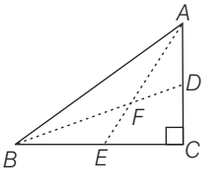
50 $EF = 3$ cm $\quad \therefore BC = 6$ cm

$FD = 4$ cm $\quad \therefore AC = 8$ cm

$AB = 10$ cm $\quad \therefore DE = 5$ cm



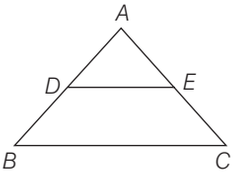
51



52 $\dfrac{AB}{AC} = \dfrac{AD}{AE}$

$\dfrac{3.6}{2.4} = \dfrac{2.1}{AE}$

$\Rightarrow \quad AE = 1.4$ cm



54 $\angle EAB = \angle OAB = 60°$

$\angle ABD = 45°$

$\therefore \quad \angle AOB = 180° - (60° + 45°) = 75°$

55 $\angle LBN = \angle BAC = \angle BCA = 70°$

$\therefore \quad \angle ABC = 180° - (70° + 70°) = 40°$

56 $\angle ECB = 60° + 90° = 150°$

$\therefore \quad \angle CEB + \angle EBC = 180° - 150° = 30°$

$\therefore \quad \angle BEC = 15° \qquad (\because \angle BEC = \angle EBC)$

$\therefore \quad \angle DEC = 60°$

$\therefore \quad \angle DEM = 60° - 15° = 45°$

57 $\angle ABQ = \angle EAY$

$\Rightarrow \quad 20 + y = 60 \Rightarrow y = 40°$

Now, $\angle BAD = 180 - (60 - 35) = 85°$

$\therefore \quad \angle ADB = 180 - (\angle BAD + \angle ABD)$

$\Rightarrow \quad \angle ADB = 180 - 105 = 75°$

58 $\angle ACB = 42° = \angle ECD$

$\therefore \quad \angle CED = 180° - (58° + 42°) = 80° = \angle FEG$

$\therefore \quad \angle FGE = 180° - (66° + 80°) = 34°$

59 $\angle BAD = 70°$

$\therefore \quad \angle BDA = 180° - (70° + 50°) = 60°$

60 $\dfrac{2x}{x} = \dfrac{3y + 8}{2y} \Rightarrow y = 8$

$\therefore \quad x = 2y = 16$

61 $\angle BCD = 27°$

$\angle BDC = 180° - (27° + 27°) = 126°$

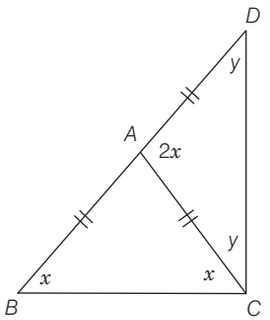
$\therefore \quad \angle ADC = 54° = \angle DAC$

$\therefore \quad \angle ACD = 180° - (54° + 54°) = 72°$

62 Let us consider that $\angle ABC = x$, therefore $\angle ACB = x$

Since $\angle CAD$ is the external angle of ΔABC, therefore, $\angle CAD = 2x$

Now, we know that $AC = AD$, therefore $\angle ACD = \angle ADC = y$



But, since sum of al the angles of $\Delta ACD = 180°$, therefore $2x + y + y = 180°$

$\Rightarrow \quad x + y = 90°$

Thus, we have $\angle BCD = \angle ACB + \angle ACD$

$\Rightarrow \quad \angle BCD = x + y = 90°$

63 By mid point theorem, we can say that

$$\frac{AE}{AC} = \frac{AD}{AB} \Rightarrow \frac{AE}{AC} = \frac{AD}{(AD + BD)}$$

$\Rightarrow \quad \dfrac{AE}{5} = \dfrac{3}{10}$

$\Rightarrow \quad AE = 1.5$

64 $\dfrac{PB}{BR} = \dfrac{AP}{AQ}$

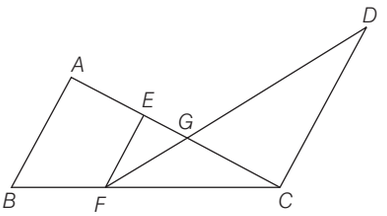
$\Rightarrow \quad \dfrac{PB}{BR} = \dfrac{2\sqrt{2}}{3\sqrt{2}} = \dfrac{2}{3}$

$\because \quad \dfrac{PB}{BR} = \dfrac{2x}{3x} \therefore BR = 6 \text{ cm}$

65 Since $EF \parallel DC$, so we have

$$\frac{EF}{DC} = \frac{EG}{CG} = \frac{FG}{DG} \Rightarrow \frac{EF}{18} = \frac{5}{10}$$

$\Rightarrow \quad EF = 9$



Now, since ΔABC and ΔEFC are similar triangles, therefore $\dfrac{AC}{AB} = \dfrac{EC}{EF}$

$\Rightarrow \quad \dfrac{AC}{15} = \dfrac{15}{9}$

$\Rightarrow \quad AC = 25$

66 Since ΔPQD and ΔABD are similar, therefore

$$\frac{QD}{BD} = \frac{PQ}{AB} \qquad \ldots(i)$$

Similarly, ΔBPQ and ΔBCD are similar, therefore

$$\frac{BQ}{BD} = \frac{PQ}{CD} \qquad \ldots(ii)$$

By adding equations (i) and (ii), we will have

$$\frac{BD}{BD} = PQ\left(\frac{1}{AB} + \frac{1}{CD}\right)$$

$\Rightarrow \quad \dfrac{1}{PQ} = \dfrac{AB + CD}{AB \times CD}$

$\Rightarrow \quad PQ = \dfrac{AB \times CD}{AB + CD} = \dfrac{xy}{x + y}$

Hence, choice (c) is the correct one.

Alternatively Consider $AB = CD = 2$, then by ladder theorem $PQ = 1$. Now substitute the values of x, y, z in the given options, you will see that only choice (d) is valid. In this question, z plays no role, so it's irrelevant.

Alternatively Apply cross Ladder Theorem, we have

$$\frac{1}{PQ} = \frac{1}{AB} + \frac{1}{CD}$$

$$= \frac{1}{x} + \frac{1}{y} = \frac{xy}{x + y}$$

Hence choice (c) is the correct one.

67 Apply similarity of triangles, $\Delta ADB \sim \Delta CDA$

$$\frac{AD}{BD} = \frac{CD}{AD}$$

$$\Rightarrow \quad \frac{3}{4} = \frac{CD}{3}$$

$$\Rightarrow \quad CD = 2.25$$

70 Since, $CF \parallel AD$, so we have,

$\angle FCA = \angle DAC = x$ and $\angle AFC = \angle MAD = x$.

Therefore, $\angle AFC = \angle FCA = x$

Thus, we can conclude that in the ΔAFC the two sides are equal.

That is $AF = AC$.

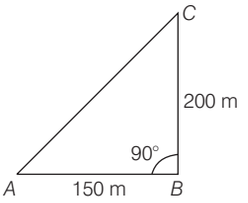
Again, since $CF \parallel AD$, so we have,

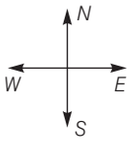
$$\frac{AB}{AF} = \frac{BD}{CD}$$

$$\Rightarrow \quad \frac{AB}{AC} = \frac{BD}{CD} \qquad (\because AF = FC)$$

71 $\Delta ABD \cong \Delta CBD$

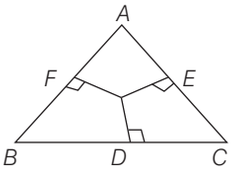
73





$$AC = \sqrt{AB^2 + BC^2}$$

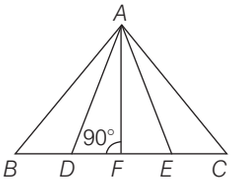
74



75 $BD = 2DF$

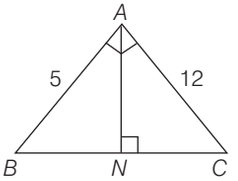
$AB^2 = BF^2 + AF^2$

and $AD^2 = DF^2 + AF^2$

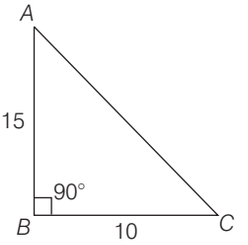


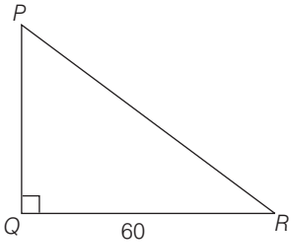
76 $$\frac{A(\Delta ANC)}{A(\Delta ANB)} = \frac{\frac{AN \times NC}{2}}{\frac{AN \times BN}{2}}$$

$$= \frac{NC}{NB} = \frac{\frac{(12 \times 12)}{13}}{\frac{5 \times 5}{13}} = \frac{144}{25}$$



77 $$\frac{AB}{BC} = \frac{PQ}{QR} \Rightarrow \frac{15}{10} = \frac{PQ}{60}$$





78 16 : 81

79 $$\frac{BC}{PQ} = \frac{AC}{AP} = \frac{AB}{AQ}$$

80 $$\frac{3}{x-3} = \frac{x-5}{3x-19} \Rightarrow x = 8 \text{ or } 9$$

81 $$h = \frac{\sqrt{3}}{2}\text{ side} = \frac{\sqrt{3}}{2} \times 2a = a\sqrt{3}$$

82 $$\frac{AB}{AC} = \frac{BD}{CD} \Rightarrow \frac{5}{AC} = \frac{2}{3} \Rightarrow AC = 7.5\text{ cm}$$

83 $$\frac{AB}{QR} = \frac{PB}{PR} \Rightarrow \frac{3}{9} = \frac{PB}{6} \Rightarrow PB = 2\text{ cm}$$

84 $$\frac{AO}{BO} = \frac{AQ}{BP}$$

$$\frac{10}{6} = \frac{AQ}{9} \Rightarrow AQ = 15\text{ cm}$$

85 $$\frac{AB}{AC} = \frac{AD}{AE} \Rightarrow \frac{12}{15} = \frac{6}{AE}$$

$$\therefore \quad AE = 7.5\text{ cm}$$

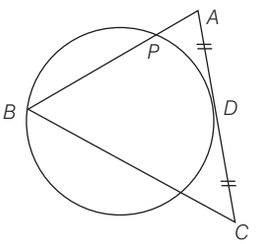
86 $AB = AC$ and $AD = CD$

$\therefore \quad AB = 2AD$

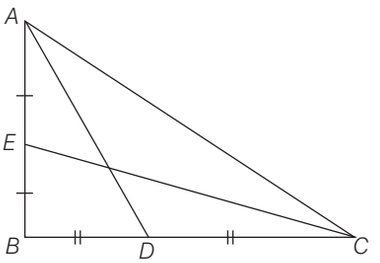
Now, since AD is a tangent

$\therefore \quad AD^2 = AP \times AB$

$$\left(\frac{AB}{2}\right)^2 = AP \times AB \Rightarrow AB = 4AP$$



87 $AC = 5\text{ cm} \Rightarrow AD = \frac{3\sqrt{5}}{2}\text{ cm}$



$AE = BE$ and $BD = CD$

$AB^2 = AC^2 - BC^2 = 25 - BC^2$ …(i)

and $AB^2 = AD^2 - BD^2 = \left(\frac{3\sqrt{5}}{2}\right)^2 - BD^2$

$= \frac{45}{4} - \frac{BC^2}{4}$ …(ii)

From eq. (i) and (ii)

$BC^2 = \frac{55}{3}$

Now, from eq. (i)

$AB^2 = 25 - \frac{55}{3} = \frac{20}{3}$

Also $CE^2 = BE^2 + BC^2 = \left(\frac{1}{2}AB\right)^2 + BC^2$

$= \frac{AB^2}{4} + BC^2 = CE^2 = \frac{5}{3} + \frac{55}{3} = \frac{60}{3} = 20$

$\therefore \quad CE = 2\sqrt{5}$ cm

88 By Apollonius theorem

$AB^2 + AC^2 = 2(AD^2 + BD^2)$ $(BD = CD)$

$100 + 196 = 2(AD^2 + 36)$

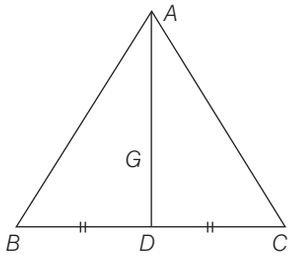
$\Rightarrow \quad AD^2 = 112$

$\therefore \quad AD = 4\sqrt{7}$

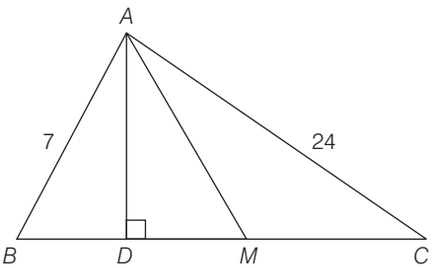
Now, since $\frac{AG}{GD} = \frac{2}{1}$

$\Rightarrow \quad AG = \frac{2}{3}AD$

$\therefore AG = \frac{2}{3} \times 4\sqrt{7} = \frac{8}{3}\sqrt{7}$ cm



89 Since AM is the median of a right angled triangle.



$\therefore \quad AM = \frac{BC}{2} = \frac{25}{2}$ and $AD = \frac{AB \times AC}{BC} = \frac{7 \times 24}{25}$

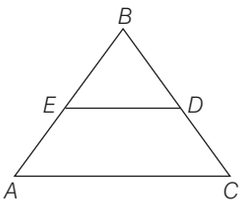
$\therefore \quad \frac{AD}{AM} = \frac{7 \times 24 \times 2}{25 \times 25} = \frac{336}{625}$

90 $\Delta BED \sim \Delta BAC$

$\therefore \quad \frac{BE}{BA} = \frac{ED}{AC} = \frac{BD}{BC} = \frac{1}{2}$

$\therefore \quad \frac{A(\Delta BED)}{A(\Delta BAC)} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$\therefore \quad A(\Delta BDE) = 7.5\,\text{cm}^2$



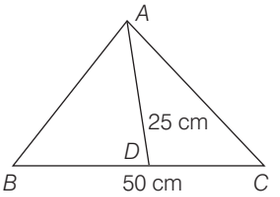
91 Apply Pythagoras theorem.

NOTE The sides given in option (d) cannot form any type of triangle.

92 $\frac{AB}{BD} = \frac{AC}{CD} \Rightarrow \frac{12}{BC + CD} = \frac{8}{CD}$

$\frac{12}{4 + CD} = \frac{8}{CD} \quad \Rightarrow \quad CD = 8$ cm

93 $AB^2 + AC^2 = 2\left[AD^2 + \left(\frac{BC}{2}\right)^2\right]$ (Appollonius theorem)



$2500 = 2\left[625 + \left(\frac{BC}{2}\right)^2\right] \Rightarrow BC = 50$ cm

94 $\angle BAC = 15°$

$\therefore \quad \angle BCA = 15°$ $(\because AB = BC)$

$\therefore \quad \angle ABC = 180° - (15° + 15°) = 150°$

$\therefore \quad \angle ABD = 30°$ $(180° - 150°)$

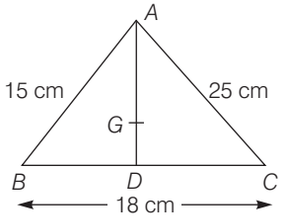
$\therefore \quad \sin 30° = \frac{AD}{AB}$

$\frac{1}{2} = \frac{AD}{AB} \quad \Rightarrow \quad AD = 5$ cm $\quad (\because AB = 10$ cm$)$

$\therefore \quad$ Area of $\Delta ABC = \frac{1}{2} \times BC \times AD$

$= \frac{1}{2} \times 10 \times 5 = 25\,\text{cm}^2$

95 $AB^2 + AC^2 = 2(AD^2 + BD^2)$



$\therefore \quad 225 + 625 = 2(AD^2 + 81)$

$\Rightarrow \quad AD^2 = 344$

$\Rightarrow \quad AD = 2\sqrt{86}$ and $GD = \frac{1}{3}AD$

$\therefore \quad GD = \frac{2}{3}\sqrt{86}$ cm

96 $\angle BDE = 115°$

$\therefore \quad \angle ADE = 65°$ and $\angle AED = 75°$

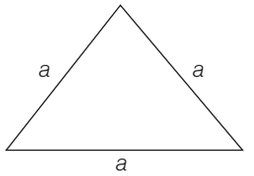
$\therefore \quad \Delta AED \sim \Delta ABC$

$\therefore \quad \frac{DE}{BC} = \frac{AE}{AB} = \frac{AD}{AC}$

$\therefore \quad \frac{2}{3} = \frac{10}{AB}$

$\Rightarrow \quad AB = 15\,\text{cm}$

97 For the given perimeter of a triangle the maximum area is enclosed by an equilateral triangle.



$\therefore \quad 3a = 24\,\text{cm}$

$\Rightarrow \quad a = 8\,\text{cm}$

$\therefore \quad \text{Area} = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times (8)^2 = 16\sqrt{3}\ \text{cm}^2$

98 $\Delta ADE \sim \Delta ACB$ (A-A-A property)

$\therefore \quad \frac{AE}{AB} = \frac{AD}{AC}$

$\frac{10}{20} = \frac{12}{AC}$

$\Rightarrow \quad AC = 24\,\text{cm}$

$EC = AC - AE = 24 - 10 = 14\,\text{cm}$

99 $\angle ACD = \angle ADC = x$

$\therefore \quad \angle CAD = (180° - 2x)$

$\angle ABC = \angle BAC = \frac{x}{2}$

$(\because \angle ABC + \angle BAC = \angle ACD = x)$

$\therefore \quad \angle BAC + \angle CAD + 81° = 180°$

$\therefore \quad \frac{x}{2} + (180° - 2x) + 81° = 180°$

$\therefore \quad \frac{3}{2}x = 81°$

$\Rightarrow \quad x = 54°$

Alternatively Applying the same process go through options.

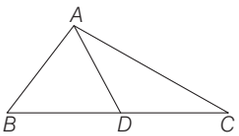
100 It is value (side) dependent *i.e.,* it is different for different triangles unlike the case of equilateral triangle.

101 Let $\angle ACB = x$

$\Rightarrow \quad \angle CAD = x$

$\Rightarrow \quad \angle ADB = 2x$

$\Rightarrow \quad \angle ABD = 2x$



But since $\angle BAC = \angle ABC$

$\Rightarrow \quad \angle BAC = 2x$

$\Rightarrow \quad \angle BAD = x$

$\therefore \angle BAC + \angle ABC + \angle ACB = 180°$

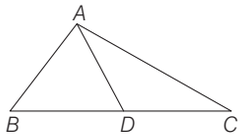
$\Rightarrow \quad 2x + x + x = 180°$

$\Rightarrow \quad x = 36°$

Hence choice (c) is the correct one.

102 As AD is the angle bisector, so

$\frac{AB}{AC} = \frac{BD}{DC}$



Further, since AD is the median, so

$BD = CD = k$ (say)

$\therefore \quad \frac{AB}{AC} = \frac{BD}{DC} = \frac{k}{k} = 1$

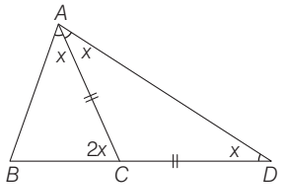
$\Rightarrow \quad AB = AC = 15\,\text{cm}$

Hence choice (c) is the correct one.

103 Let $\angle BAC = x$, then $\angle ACB = 2x$. Now, extend BC till a point D such that when A and D are connected $\angle CAD$ becomes equal to $\angle BAC$.

But since $\angle ADC + \angle DAC = \angle ACB$

$\Rightarrow \quad \angle ADC = x$



Therefore, $AC = DC = 4$ cm. And,

so $BD = BC + CD = 10$ cm

Now you can see that ΔBAC and ΔBDA are similar triangles.

Therefore,

$\frac{AB}{BC} = \frac{BD}{AB}$

$\Rightarrow \quad \frac{AB}{6} = \frac{10}{AB}$

$\Rightarrow \quad AB = \sqrt{60} = 2\sqrt{15}\,\text{cm}$

Hence choice (d) is the correct one.

104 Since BD is the angle bisector of $\angle ABC$, therefore

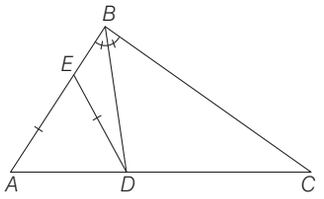
$\frac{AB}{BC} = \frac{AD}{CD} \Leftrightarrow AD = \frac{(AB)(CD)}{BC}$

But, it is given that

$\frac{(BD)(BC)}{CD} = \frac{(BE)(AB)}{AE}$

$\Rightarrow \quad \frac{(BD)(BC)}{(CD)(AB)} = \frac{BE}{AE}$

$$\Rightarrow \quad \frac{BD}{AD} = \frac{BE}{AE}$$



It implies that DE is the angle bisector of $\angle ADB$.

$\therefore \quad \angle BDE = \angle EDA = \angle EAD = x°$ (say)

$\therefore \quad \angle BDA + \angle DAB + \angle ABD = 2x + x + 66 = 180°$

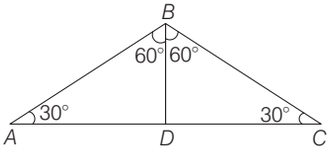
$\Rightarrow \quad x = 38°$

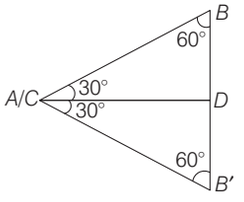
$\Rightarrow \quad \angle BAD = 38°$ and $\angle ACB = 28°$

$\therefore \quad \angle BAD - \angle ACB = 10°$

Hence choice (a) is the correct one.

105 Consider the following figures. If you cut the first figure from the mid-point of its base along its altitude BD and then rotate it clockwise about the point D by 180°, then you will get an equilateral triangle, in which points A and C will overlap. Now it would become very easy to find the area of the given triangle as even though the shape of the triangle changes but its area remains the same.





The area of the equilateral triangle $= \frac{\sqrt{3}}{4} \times a^2 = \frac{\sqrt{3}}{4}$

$(\because a = 1)$

Hence choice (a) is the correct one.

Alternatively Consider the first figure. Since the triangle is an isosceles one, so the sides AB and AC would be of same length. And so the two angles BAC and BCA would be 30° each.

As the triangle ABC is an isosceles one, so the perpendicular dropped from the angle B on the opposite side AC will bisect the side AC as well as the angle ABC.

Then there will be two congruent triangles ABD and CBD. Now using 30-60-90 degree theorem (or trigonometric ratios) we can find the base $AD = DC = \frac{\sqrt{3}}{2}$ and altitude $BD = \frac{1}{2}$.

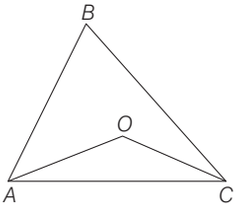
Therefore area of $\Delta ABC = \frac{1}{2} \times AC \times BD$

$$= \frac{1}{2} \times \sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{4}$$

Alternatively Consider the first figure. Using 30-30-120 degree theorem or Cosine rule/Sine rule, we will have $AC = \sqrt{3}$ unit. After this we can use the formula for area of a triangle $\sqrt{s(s-a)(s-b)(s-c)}$, where s is the semiperimeter of the triangle and a, b, c are the sides of it.

106 $\angle AOC = 180 - \frac{1}{2}(\angle BAC + \angle BCA)$

$\Rightarrow \quad (\angle BAC + \angle BCA) = 112°$

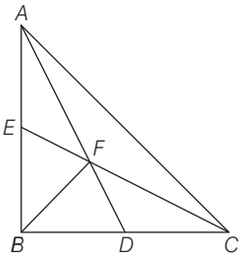


and $\quad \angle ABC = 180° - (\angle BAC + \angle BCA)$

$\Rightarrow \quad \angle ABC = 180° - 112 = 68°$

Hence choice (c) is the correct one.

107 First of all, let's connect B to F. Now, we can easily figure out that the area of $\Delta BFD = \Delta CFD = x$, as the base and height of these two triangles are same. Similarly, we can find out that area of $\Delta AEF = \Delta BFE = y$.



But, we know that area of $\Delta ABD = \Delta EBC$.

Therefore, $2x + y = x + 2y$

$\Rightarrow \quad x = y$

Therefore, area of $\Delta AEF = x$ and area of quadrilateral $BEFD = 2x$.

Hence choice (c) is the correct one.